

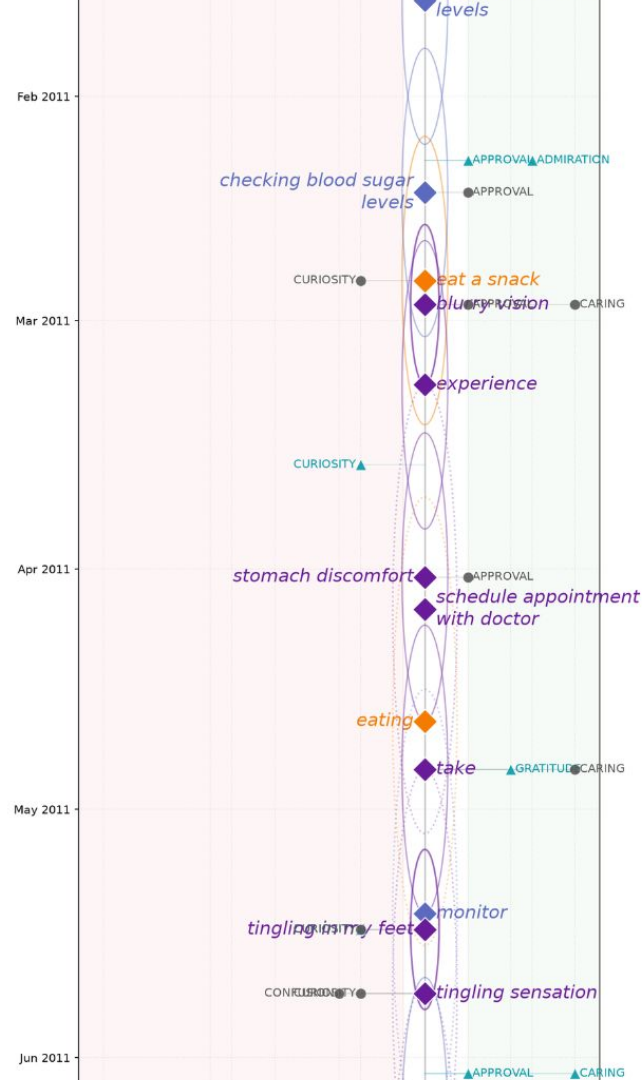
DELPH-IN Summit 2026

Event-based Knowledge Graphs with the ERG

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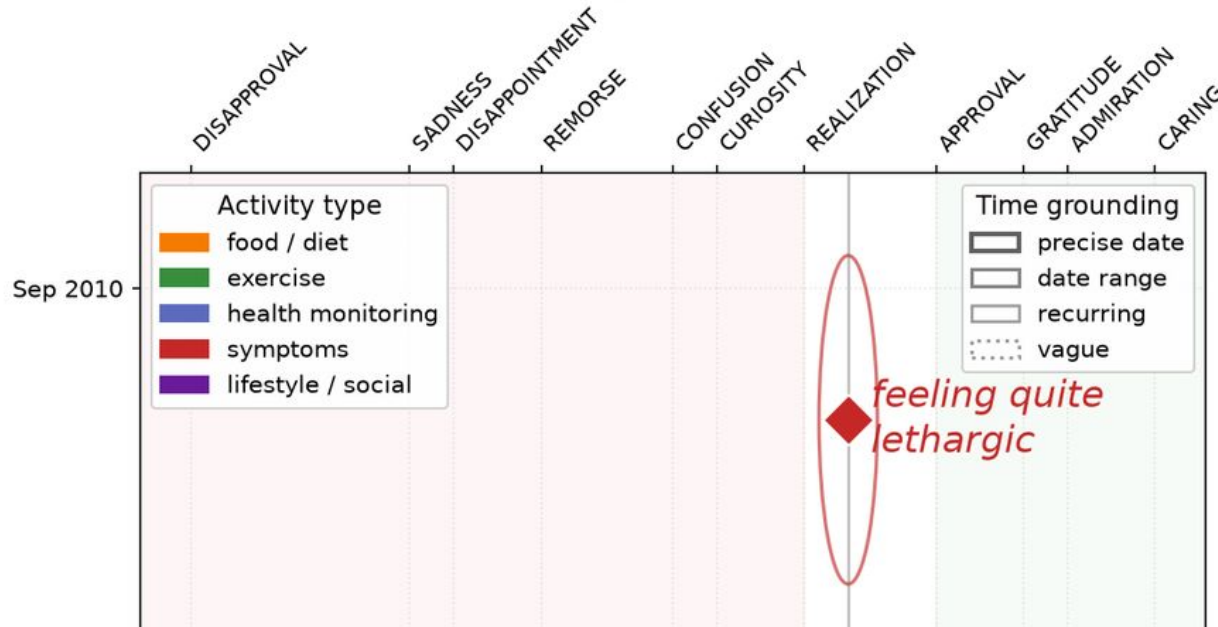
Summary

- Collaboration with TNO (Dutch “Applied Scientific Research”)
- Medical NLP: **Patient Monitoring (Diabetes use case)**
- Uses Leolani-KG platform
 - **GRaSP: Grounded Representation and Source Perspective**
 - Model **Carer-Patient** interactions with Lifestyle Coaches
 - Focus on **representing Activities of Daily Life as a timeline**
- Extracting (SRL-style) event-based information to build a KG
 - LLMs are the easy choice, **ERG is the interesting choice**

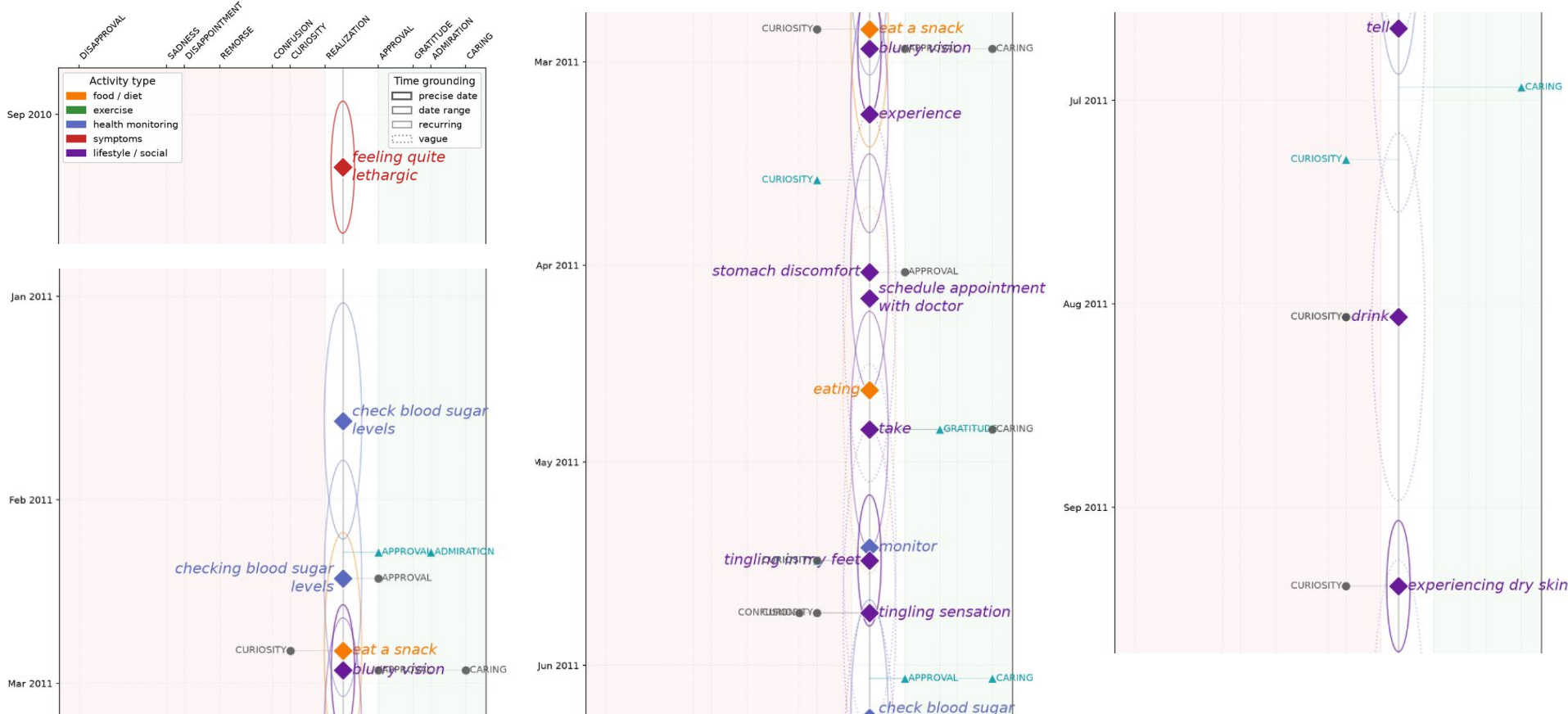
Goal - Monitoring Timeline

Activity Timeline — Emotion Perspectives by Speaker

◆ activity (coloured by type) ○ = Rudolf △ = agent oval = grounded time (size encodes precision)
Emotion



Goal - Monitoring Timeline



Data

- Unfortunately, we only have LLM-generated
 - Expert-validated, but clearly different (longer, etc.)
- Data:
 - 16 demographically and medically diverse diabetic personas
 - 256 multi-turn conversations
 - 2,027 utterances, 4,833 sentences

Data Example



LLM generated
data incoming!

Data Example

```
{  "chat": 0,  "human": "Jan",  "date": "2013, Apr, 13",  "turns": [  
    { "turn": 1, "speaker": "agent", "utterance": "Hey Jan, I see that you're a fan of cycling. Has the Dutch weather  
been favorable for it lately?" },  
    { "turn": 2, "speaker": "Jan", "utterance": "It's been quite unpredictable recently, so I haven't been out as often  
as I'd like. How does this affect my diabetes management?" },  
    { "turn": 3, "speaker": "agent", "utterance": "Physical activity, like cycling, is crucial in managing your diabetes.  
It helps your body use insulin better and can lower your blood sugar levels. If weather conditions are interfering  
with your routine, perhaps you could look for indoor exercises to stay active." },  
    { "turn": 4, "speaker": "Jan", "utterance": "Any suggestions for indoor exercises that could suit me?" },  
    { "turn": 5, "speaker": "agent", "utterance": "Definitely, Jan. Indoor walking or marching in place can be an  
excellent start. If possible, you could consider investing in a stationary exercise bike, which would be similar to your  
outdoor cycling. Strength training exercises using light weights or resistance bands are also beneficial." },  
    (more turns with Jan)  
  ] } (... more chats with the same and other people)
```

ERG over the data

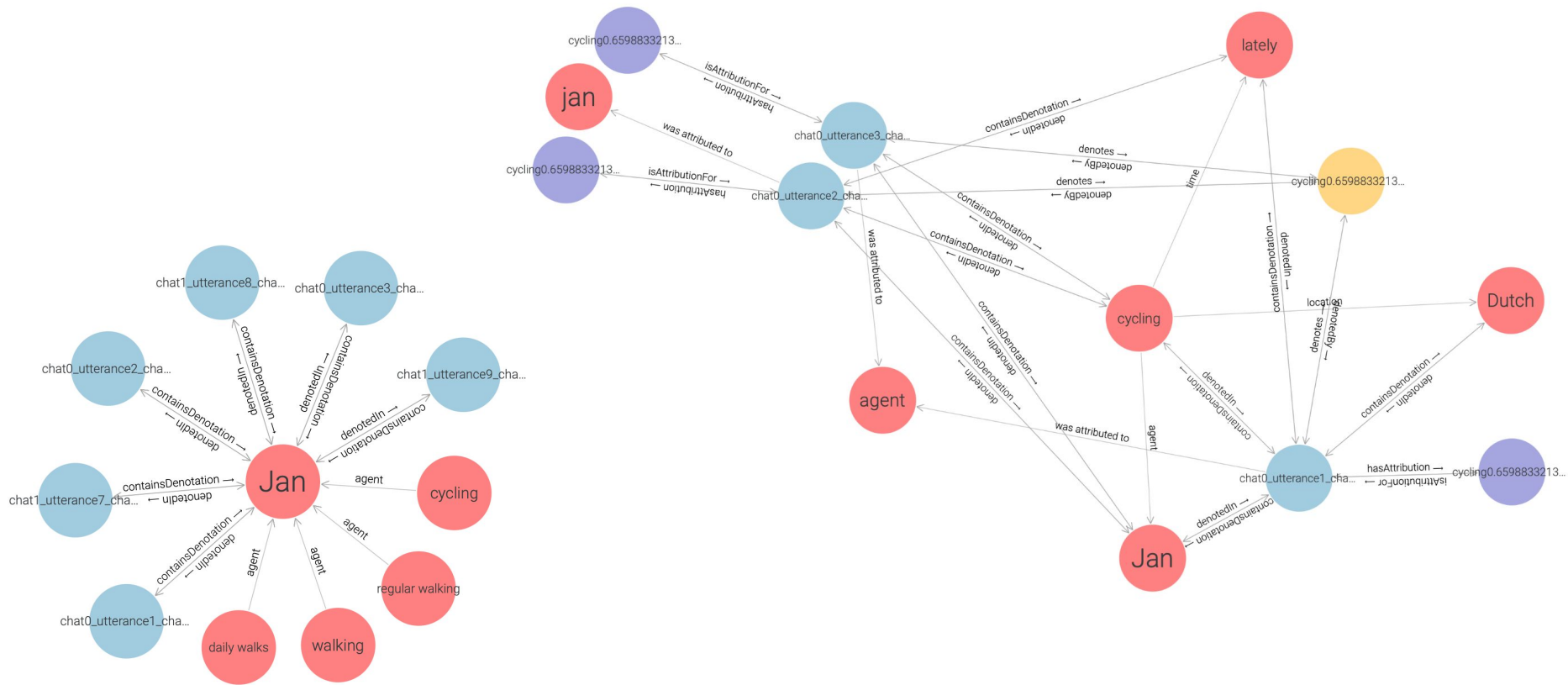
- ERG parses 98.59% of the data (non-robust grammar)
- “Failures” seem like they are mostly features, not bugs...

Showing 1 to 5 of 68 entries (filtered from 4,833 total entries)

Previous 1 2 3 4 5 ... 14 Next

id	wf	readings			length	input	
		old	new	Δ			
178	1	0	0	0	22	It's crucial then to focus on bringing those levels down, Jan. You might want to review your diet, exercise, and medication routine.	PARSE
191	1	0	0	0	12	Yes, Jan. Thirst and dry mouth can indicate high blood sugar levels.	PARSE
229	1	0	0	0	36	While in most cases, exercise lowers blood sugar levels by increasing your body's insulin sensitivity, intense or prolonged physical activity can sometimes cause your body to produce stress hormones, which can raise blood sugar levels temporarily.	PARSE
270	1	0	0	0	12	I'll bear that in mind and try to adjust my dinnertime accordingly.	PARSE
411	1	0	0	0	8	Hydration complements your balanced diet, not substitute it.	PARSE

GRaSP-based Knowledge Graphs



Vibe-Coded Annotation Bench

Turn 1 **agent**

Hey Jan, I see that you're a fan of cycling. Has the Dutch weather been favorable for it lately?

chat0.1 **cycling** [exercise] by piek

EMOTION **curiosity**

CERTAINTY **uncertain**

FACTUALITY **expect**

AGENT

PATIENT

AGENT_PATIENT

EXPERIENCER

PARTICIPANT

QUALIFICATION

INSTRUMENT

LOCATION

RESULT

TIME

Jan (person) ✕

lately ✕

lately: range 2013-03-13→2013-04-13 ✕

+ Role

+ Time Resolved

Edit Activity

Delete

+ Add Activity Annotation

or select text in the utterance to assign a role

Turn 2 **Jan**

It's been quite unpredictable recently, so I haven't been out as often as I'd like. How does this affect my diabetes management?

chat0.1 ↗ refers to previous activity as **been out** [exercise] by piek

EMOTION **disappointment**

CERTAINTY **neutral**

FACTUALITY **deny**

AGENT

PATIENT

AGENT_PATIENT

EXPERIENCER

PARTICIPANT

QUALIFICATION

INSTRUMENT

LOCATION

RESULT

TIME

as often as I'd like ✕

as often as I'd like: recurring (often) ✕

+ Role

+ Time Resolved

Edit Activity

Delete

+ Add Activity Annotation

or select text in the utterance to assign a role

What will happen?

- Paper Plan: annotate dataset >> test multiple triple-extract methods
 - LLMs are inconsistent and hallucinate (still: the best we have)
 - ERG would be more consistent but will lack some bits of the pipeline needs (e.g., coreference resolution)
 - “We” don’t know what kind of predicates we want (closer or further from the surface form)
- Personal Goal: try to produce triples from ERG produced semantics
 - I will probably need to use LLMs to link things – limit errors to poor links and not to made up things
- Any notes on similar/useful work would be greatly appreciated

Thank you!



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